

Anti-parasitic Drugs

Mechanisms, Clinical Use, and Drug Selection

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Learning Objectives

1. Classify antiparasitic drugs based on parasite type
2. Understand key mechanisms of action
3. Select appropriate therapy for common infections
4. Recognize major adverse effects and limitations
5. Apply clinical patterns to select appropriate therapy

البروتوزوا

الكائنات وحيدة الخلية

طرق انتقال الديدان الطفيلية!!

HELMINTHS

الديدان الطفيلية

Ectoparasites

الطفيليات الخارجية

مثل القمل

How Do We Approach Antiparasitic Therapy?

Effective treatment requires:

- Identification of the type of parasite
 - Consideration of site of infection (luminal vs tissue)
 - Mechanism of action guides drug selection
- (Location of parasite (e.g., luminal vs tissue) determines drug choice)



Antiparasitic drugs are **organism-specific**, unlike broad-spectrum antibiotics

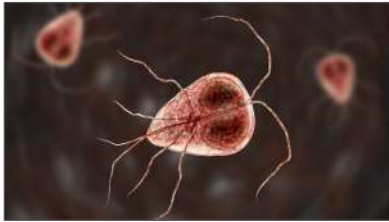
Classification

Category	Organisms	Key Drugs
Protozoa	Giardia, Entamoeba, Plasmodium	Metronidazole, Artemisinin
Helminths	Nematodes, Cestodes, Trematodes	Albendazole, Ivermectin, Praziquantel
Ectoparasites	Scabies, lice	Permethrin, Ivermectin

PROTOZOA

الأولي

Unicellular organisms causing intestinal and systemic infections

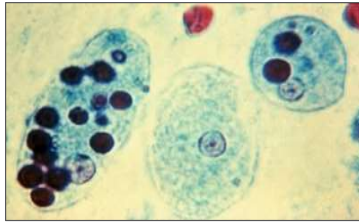


Giardia

(jee-AR-dee-uh)

Giardiasis

(jee-ar-DYE-uh-sis)



Entamoeba

(en-tuh-MEE-buh)

Amoebiasis

(am-uh-BYE-uh-sis)



Plasmodium

(plaz·mow·dee·uhm)

Malaria

(muh-LAIR-ee-uh)

Protozoal Infections

- Unicellular organisms
- **Infections:**
 - Gastrointestinal tract (Giardia, Entamoeba)
 - Blood (Plasmodium)

- **Treatment depends on:**

Luminal vs. tissue (invasive) infection

- Luminal drugs: eradicate parasites in the gut
- Tissue drugs: treat invasive disease
- For some infections, you need both drugs:

Tissue drug (Metronidazole) + luminal drug (e.g., Paromomycin)

Type	Definition	Treatment
Luminal	Parasite in intestine only	Luminal drugs
Tissue	Parasite invades tissues	Systemic drugs

★ اهم ستي يعرف
ذي الدتيار

Metronidazole

- MOA: Free radical formation - DNA damage.
- PK: Excellent tissue penetration. It is a "tissue drug" so it often needs a luminal agent to kill cysts in the gut.
- Uses: Amoebiasis & Giardiasis.
- Adverse Effects: Metallic taste & Peripheral neuropathy

Mechanism of Action (MOA):

- A prodrug, gets activated in anaerobic organisms
- Reduced in anaerobic organisms → forms free radicals
→ DNA damage → cell death
- Does NOT reliably eradicate luminal cysts
May require luminal agent

Pharmacokinetics:

- Excellent tissue penetration (treats invasive disease)
- Available as oral and IV

Disulfiram is a drug used to stop alcohol addiction. It prevents the body from breaking down alcohol, which causes severe sickness (like vomiting and headache) if someone drinks. **Metronidazole** does the exact same thing, which is why it's called a **Disulfiram-like reaction**.

A treatment for:

- Giardia.
- Entamoeba.
- Anaerobic bacteria

Clinical Use:

1. Giardiasis
2. Amoebiasis (tissue infection)
3. Often combined with luminal agents

Adverse Effects:

1. Metallic taste
2. Nausea, vomiting
3. Disulfiram-like reaction with alcohol
4. Peripheral neuropathy (prolonged use)

Metabolic Toxic Neuropathy

Antimalarial Drugs

Classification:

Artemisinin (Artemether), Quinolines (Chloroquine), Antifolates (Pyrimethamine)

- Resistance patterns influence drug choice

Class	Artemisinin	Quinolines	Antifolates
MOA	Free radical formation: ✱✱ Activated by iron → Produces free radicals → Damages parasite proteins and membranes	Inhibits heme detoxification: ▪ Inhibits conversion of toxic heme → hemozoin (non-toxic) ▪ Leads to parasite toxicity	Inhibits folate synthesis: Inhibiting the folate biosynthetic pathway in the parasite (essential for DNA synthesis and cell replication)
Use	First-line for Plasmodium falciparum ✱✱	Ranges from treating active blood-stage infections to preventing relapses and transmission	Prophylaxis (in pregnancy, infants, travelers)
S.E	Generally, well tolerated	Retinopathy (long-term)	▪ Bone marrow suppression ▪ High-dose folic acid supplementation can reduce its efficiency مايجير منه تاثيره الدواء

Pattern Recognition (PROTOZOA)

EXAM

IMPORTANT

Greasy = Giardia
Bloody = Entamoeba
Cyclic fever = Malaria

Table is for illustration (know what is important)

Feature	Giardia	Entamoeba	Malaria
Stool	Greasy (foul-smelling)	Bloody	None
Fever	No	Mild	High
Location	Intestine	Intestine + tissue	Blood (RBC)
Key symptom	Bloating	Amoebic Dysentery	Cyclic fever
Medication	Metronidazole	Metronidazole + luminal agent	Artemisinin-based combination therapy (ACT)

Note: blood → think invasion, not just luminal infection

Q1: A traveler presents with **bloating** and **greasy, foul-smelling stool**. What is the diagnosis?

A: Giardia

Q2: A patient presents with **mild fever** and **bloody stool** (Dysentery). What is the diagnosis?

A: Entamoeba

Q3: A patient presents with **high cyclic fever** but has **no stool symptoms**. What is the diagnosis?

A: Malaria

HELMINTHS

الديدان

Multicellular parasites affecting GI tract and tissues

الطفيلية



Roundworm

المستد

يرة



Tapeworm

الشر

يطية



Flukeworm

المفلطح

ة

Helminths Infections

- Multicellular parasites

Types:

1. Nematodes (roundworms, pinworm)
2. Cestodes (tapeworms)
3. Trematodes (flukes)



Names origin is Greek:

- *Nema* = threads. *Odes* = like
- *Cestus* = belt
- *Trema* = hole (refer to sucker pores)

فهم

Medication Choice

Location

For helminths, think where **the worm lives** in the body.
(**not** luminal vs tissue)

Intestinal worms	Tissue worms
Stay in gut	Liver, lungs, blood
Albendazole works well ✨ ✨ ✨	Need drugs that reach tissues. Example: ▪ Echinococcus → liver cysts → Albendazole ✨ ✨ ✨ ▪ Schistosoma → blood vessels → Praziquantel

Albendazole / Mebendazole

Albendazole basically
STARVES the parasite
to death



MOA:

- Broad-spectrum anti-helminthic
(covers most intestinal worms)
- Inhibits microtubule formation → prevent the parasite from absorbing glucose → energy depletion (↓ ATP) → parasite death

Pharmacokinetics:

- Distribute well throughout body tissues (brain, liver, lungs, and muscle)
- Absorption enhanced with a fatty meal

Clinical Use: (Albendazole)

▪ Broad-spectrum:

1. Enterobiasis (pinworm)
2. Ascariasis
3. Hookworm

▪ Also used in:

1. Hydatid disease (a tapeworm disease, forms slow-growing, fluid-filled cysts, in the liver and lungs)

Adverse Effects: (Albendazole)

1. GI upset
2. Elevated liver enzymes
3. Bone marrow suppression (rare)

Helminths Infections (other medications)

Ivermectin

MOA:

- Enhances GABA-mediated chloride influx
- Causes paralysis of parasite

Clinical Use:

Strongyloidiasis (داء الأسطوانيات)

(stron-jih-LOY-deez)

Praziquantel

MOA:

- Increases calcium permeability
- Leads to muscle contraction and paralysis

Clinical Use:

Schistosomiasis (داء البلهارسيا)

(shis-toh-soh-MY-uh-sis)

Pattern Recognition (Helminths)

EXAM

اول ثلاثة اهم شي

- Peri-anal itching
- Anemia
- Cysts

Clinical Clue	Likely Parasite Type	Key Example	First-line Drug
<u>Peri-anal itching</u> (at night)	<u>Nematode</u>	Enterobius (pinworm)	Albendazole
<u>Iron deficiency anemia</u>	Nematode	<u>Hookworm</u>	Albendazole
<u>Visible worm in stool</u>	Nematode	<u>Ascaris</u>	Albendazole
<u>Cysts</u> in liver/lung	Cestode	Echinococcus	Albendazole
<u>Blood in urine</u> (endemic area)	Trematode	Schistosoma	<u>Praziquantel</u>

Enterobiasis (en-tuh-roh-BIGH-uh-siss)

Ascariasis (as-kuh-RIE-uh-sis)

Echinococcus (i-KYE-nuh-KAH-kuhs)

Schistosomiasis (shis-toh-soh-MY-uh-sis)

ECTOPARASITES

الطفيليات الخارجية

Parasites living on skin surface causing itching and inflammation



Scabies Mites

عث الجرب



Lice

قمل



Tick

قرادة

Ectoparasites (Overview)

- Live on skin surface
- Cause itching and inflammation ✖ ✖
- Transmission is usually via close contact

Medication Choice

Location → skin surface, so treatment is:

1. Topical first → Permethrin ✖ ✖
2. Systemic (only if severe) → Ivermectin ✖ ✖

الدواء هنا مهم
اكثر من لمن انذكر
فوق

Ticks (clinical differentiation point only) ✖ MSQ

- Attach to skin and feed on blood
- Managed by mechanical removal (**not antiparasitic drugs**)
- Important due to disease transmission

Treatment

- Permethrin

MOA: Disrupts sodium channels → neurotoxicity in parasites

Uses: First-line for scabies and lice

- Ivermectin

- Used in severe or resistant cases
- Oral option for widespread infestation

Q: A person went into the forest and after leaving, discovered a tick attached to their skin. What is the best management?

A: Physical removal (mechanical removal), making sure to remove the head

Pattern Recognition (Ectoparasites)

EXAM

Minimal itching
Disease transmission
Tick removal

Feature	Scabies	Lice	Ticks
Location	Inside skin	On hair	Attached to skin
Itching	Severe, at night	Mild-moderate	Minimal
Key finding	Burrows	Nits (eggs)	Visible tick
Risk	Skin infestation	Hair infestation	Disease transmission
Treatment	Topical Permethrin +/- oral Ivermectin	Permethrin (topical)	Physical removal of the tick + monitoring (symptoms)
Image	 Skin Burrows	 Lice Nits	 Tick Removal

References

Q1: A patient presents with severe itching that gets worse at night. The exam shows small burrows inside the skin. What is the diagnosis and treatment?

A: Scabies. Treated with **Topical Permethrin.**

Q2: A child complains of head itching. Small nits (eggs) are found attached on the hair. What is the diagnosis and treatment?

A: Lice. Treated with **Permethrin (topical).**

1. Lippincott's Illustrated Reviews. Pharmacology
2. Katzung's Basic and Clinical Pharmacology
3. Goodman & Gilman's The Pharmacological Basis of Therapeutics

TRANSLATED AS:

It referred to a person who habitually ate at someone else's expense !

It became a biological term in the 16th century.

G
R
E
E
K

ANCIENT

es, alongside"
-sitos "grain, food".

"one who eats at
another's table"

